

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE –
RAIGAD -402 103

Winter Semester Examination – December - 2019

Branch: Information Technology

Subject with Subject Code: - Graph Theory (BTITOE505A)

Date:- 18/12/2019

Sem.:- IV

Marks: 60

Time:- 3 Hr.

Instructions to the Students

1. Each question carries 12 marks.
2. Attempt **any five** questions of the following.
3. Illustrate your answers with neat sketches, diagram etc., wherever necessary.
4. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly

- | | (Marks) |
|---|----------------|
| Q.1. (a) State Handshaking theorem for undirected graph and then prove that an undirected graph has even number vertices of odd degree. | [6] |
| (b) If G is a bipartite graph and the bipartition of G is X and Y , then show that:
$\sum_{v \in X} \deg(v) = \sum_{v \in Y} \deg(v)$ | [6] |
| Q.2. (a) Prove that: If G is a graph with at least one cut-vertex, then at least two of the blocks of G contain exactly one cut vertex. | [6] |
| (b) What is an automorphism of graphs? Give the automorphism for a K_n graph. | [6] |
| Q.3. (a) Prove that a graph G is tree if and only if G is acyclic and $ E(G) = V(G) - 1$ | [6] |
| (b) State and explain Menger's theorem with the help of example. | [6] |
| Q.4. (a) Prove that a connected graph G is Eulerian if and only if every vertex of G has even degree. | [6] |
| (b) Prove that if graph G is Hamiltonian, then for every nonempty subset: $S \subseteq V(G)$, $C(G-S) \leq S $. | [6] |
| Q.5. (a) Explain planar graph and show that if G is connected planar graph with n vertices, m edges and r regions, then $n-m+r=2$ | [6] |
| (b) State Kuratowski's theorem for non-planar graphs. find chromatic number of C_n and $K_{m,n}$ graphs. | [6] |
| Q.6. (a) Explain Perfect Matching in graph with example. | [6] |
| Explain Ramsay's theorem for monochromatic cliques. | [6] |

End of Paper